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{
  "id": "SOP-IBM3090-001",
  "title": "IBM 3090 Processor Complex: Machine Check (MCH) Analysis & Recovery",
  "system": "IBM 3090",
  "component": "3090 CPU / TCM (Thermal Conduction Module)",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Complex Systems Recovery (GG24-3346)",
  "description": "Diagnostic procedure for analyzing Machine Checks and TCM logic failures on the
IBM 3090 system.",
  "steps": [
    {
      "step": 1,
      "action": "Machine Check Interruption Code (MCIC) Audit",
      "details": "When an MCH occurs, retrieve the MCIC from storage locations 232-239. Check
for 'Instruction Processing Damage' or 'System Damage'.",
      "check": "Bit 0 (System Damage) indicates a non-recoverable error."
    },
    {
      "step": 2,
      "action": "TCM Isolation",
      "details": "The 3090 uses high-density TCM modules. Use the Service Processor to read
the Logout Area. Identify the failing TCM address.",
      "indicator": "Check the power/coolant levels for the identified TCM to rule out
environmental failure."
    },
    {
      "step": 3,
      "action": "Run Maintenance Microdiagnostics",
      "details": "Execute the 'System Functional Tests' (SFT) from the console. These tests
specifically stress the dyadic processor paths.",
      "logic": "If the error is recoverable (SMC - Soft Machine Check), verify if the
'Instruction Retry' hardware successfully re-executed the failing cycle."
    },
    {
      "step": 4,
      "action": "Module Level Repair",
      "details": "The diagnostics will output a FRU (Field Replaceable Unit) code. Cross-
reference this with the IBM 3090 Parts Catalog.",
      "remedy": "Replace the failing TCM or the associated channel control board."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Machine Check light solid, System Damage ON",
      "next": "Isolate failing TCM via SVP logout area."
    },
    {
      "condition": "Intermittent check during heavy I/O",
      "next": "Check Channel-to-TCM interface logic boards."
    },
    {
      "condition": "Retry logic success",
      "next": "Log error as Soft Machine Check. No immediate repair needed."
    }
  ]
}

```